CLASS 9 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. An oxide of nitrogen contains 63.6 per cent nitrogen. The empirical formula (N = 14, O = 16) is: 1

(a) N_2O_3 **(b)** NO **(c)** N_2O **(d)** NO_2

2. The atomic number of an element is 17 and its ion has a charge of -1. The ion's configuration is the same as that of: 1

(a) helium **(b)** krypton **(c)** neon **(d)** argon

3. An experimental result is 4.8 and the model gives 5.0. The first step in an honest investigation is to: 1

(a) check the experiment and the assumptions in the model
(b) change the 4.8 to 5.0
(c) drop the experiment
(d) ignore the model

4. A student mixes sand and powdered sugar. The best method of separation is: 1

(a) sublimation **(b)** dissolve in water, filter, evaporate
(c) magnet **(d)** winnowing

5. The simplest ratio of atoms in a compound with 40 per cent C, 6.7 per cent H, 53.3 per cent O is: 1

(a) 2 : 1 : 1 **(b)** 1 : 1 : 1 **(c)** 1 : 2 : 1 **(d)** 1 : 2 : 2

6. A radioactive isotope loses half of its atoms every 5 days. The fraction remaining after 15 days is: 1

(a) 1/8 **(b)** 1/16 **(c)** 1/2 **(d)** 1/4

7. Assertion (A): A measurement can never be exactly correct. Reason (R): Every instrument has a least count and every observation has some uncertainty. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. Assertion (A): The solute in a copper sulphate solution cannot be separated by filtration. Reason (R): The solute particles are much smaller than the pores of the filter paper. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. Assertion (A): Carbon tetrachloride has a lower melting point than sodium chloride. Reason (R): It is a covalent compound with weak forces between molecules. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): Rutherford's model could not explain the stability of the atom. Reason (R): According to classical physics an accelerating electron radiates energy and would spiral into the nucleus. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. A cube of side 2.0 cm (+/- 0.1 cm). The percentage error in the volume is about: 1

(a) 10 per cent **(b)** 1.5 per cent **(c)** 15 per cent **(d)** 5 per cent

12. In a chromatogram of an ink with R_f values 0.3, 0.6, 0.9, the component most soluble in the solvent is: 1

(a) the one with R_f 0.6 **(b)** the one with R_f 0.9
(c) the one with R_f 0.3 **(d)** all equal

13. The formula unit mass of the compound $\mathrm{Ca_3(PO_4)_2}$ is (Ca = 40, P = 31, O = 16): 1

(a) 262 u **(b)** 278 u **(c)** 310 u **(d)** 230 u

14. An element has mass number 31 and 16 neutrons. The number of valence electrons is: 1

(a) 5 **(b)** 15 **(c)** 3 **(d)** 8

15. Which of the following is a good reason for conducting a pilot experiment? 1

(a) to produce final results
(b) to save time by skipping the main experiment
(c) to test the procedure and find problems
(d) to hide errors

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. How many formula units are in 117 g of NaCl relative to 58.5 g? 2

17. State two reasons why Thomson's model could not explain Rutherford's results. 2

18. A student's data for g from five trials are 9.7, 9.9, 9.8, 9.6, 9.8 m/s^2. Find the mean and the range. 2

19. Why is it possible for a solution to be both dilute and saturated? Give an example. 2

20. An oxide of copper contains 80 per cent copper by mass. Find its empirical formula (Cu = 64, O = 16). 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. On heating 24.5 g of potassium chlorate, 14.9 g of potassium chloride is left. Find the mass of oxygen liberated and the percentage of oxygen in the original sample. 3

22. Explain, using Bohr's idea, why the elements in the same group show similar chemistry and the elements in the same period show a gradual change. 3

23. Estimate the mass of air in a classroom of size 8 m x 6 m x 3 m (density of air 1.2 kg per cubic metre). 3

24. Explain how forensic scientists use chromatography to identify ink used in a forged document. 3

OR

The total mass of reactants in a closed container was 50 g. After the reaction the mass of the products found was 49.8 g. Suggest two reasons and comment on the law. 3

25. Explain the use of uranium-235 in nuclear reactors and the safety concerns. 3

OR

Explain how science and ethics are linked using the example of testing medicines. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Discuss the strengths and limitations of the scientific method with five points and two examples of discoveries involving chance and persistence. 5

OR

Explain how the principle of chromatography, the concept of R_f and the effect of the solvent choice may be used to identify unknown dyes in food colours. 5

27. Explain how Dalton's atomic theory accounted for the laws of chemical combination and which postulates failed later (subatomic particles, isotopes, isobars) and which ideas remain valid. 5

OR

Explain, with reference to the gold foil experiment, how the nuclear radius can be estimated and why the nucleus is so dense (nuclear radius about 10^-15 m, mass about 10^-26 kg). 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND